

IR Transmission Research for Multi-Transmitter Systems

Abstract

This research investigates infrared (IR) transmission in a multi-transmitter system using the NEC communication protocol. The system operates with a 38 kHz carrier and 940 nm infrared radiation. Interference was observed when multiple transmitters were placed close to one another, motivating investigation of transmitter directivity, optical coverage, and isolation.

The receiver system consists of a 12×12 array of individual IR receivers occupying a total area of approximately 900 cm^2 , corresponding to a $30 \text{ cm} \times 30 \text{ cm}$ receiving surface. The expected transmitter-to-receiver separation is approximately 1–2 m.

Testing with a KY-005 transmitter showed insufficient optical coverage for reliable reception across the complete receiver array. A directional 940 nm transmitter using Vishay TSAL6200 infrared emitters is therefore proposed. The transmitter uses two series-connected IR LEDs as its basic optical source and can be expanded to four LEDs using a second independently current-limited branch.

An AO3400A logic-level N-channel MOSFET is used to switch the emitters from the 3.3 V FPGA control signal. The design emphasizes optical directivity and efficient use of emitted infrared power rather than simply increasing electrical power. The resulting architecture also provides a platform for experimentally studying the relationship between beam angle, transmission distance, receiver coverage, and interference between adjacent transmitters.

1 Introduction

During development of a parallel NEC infrared transmission system, interference was observed when multiple transmitters were positioned in close proximity. Because the transmitters operate using the same infrared wavelength and carrier frequency, radiation from neighboring transmitters can reach unintended receivers and interfere with the desired signal.

The transmitter therefore has two competing requirements:

- sufficient infrared intensity to achieve reliable communication across the complete receiver array at a distance of approximately 1–2 m;
- sufficiently controlled optical radiation to minimize illumination of neighboring receiver regions.

The design presented in this work addresses these requirements by using directional 940 nm infrared emitters and a dedicated MOSFET switching circuit.

The transmitter PCB is designed to support multiple IR LEDs so that the required optical output can be determined experimentally while maintaining controlled beam geometry.

2 Signal Characteristics

The communication signal follows the NEC infrared protocol and uses a carrier frequency of

$$f_c = 38 \text{ kHz.}$$

The corresponding carrier period is

$$T_c = \frac{1}{38\,000} \approx 26.3 \mu\text{s.}$$

The selected infrared emitters operate at a peak wavelength of

$$\lambda_p = 940 \text{ nm,}$$

which lies within the near-infrared (NIR) region of the electromagnetic spectrum.

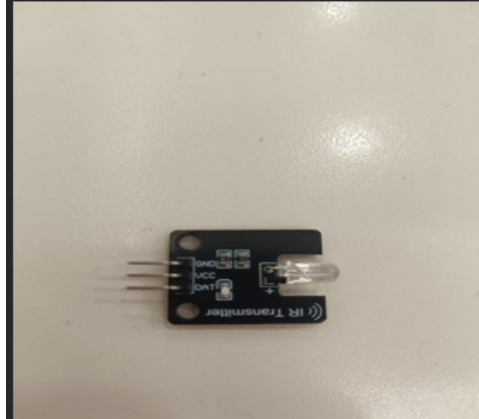


Figure 1: Infrared transmitter element

The location of near-infrared radiation within the electromagnetic spectrum is illustrated in Figure [2](#).

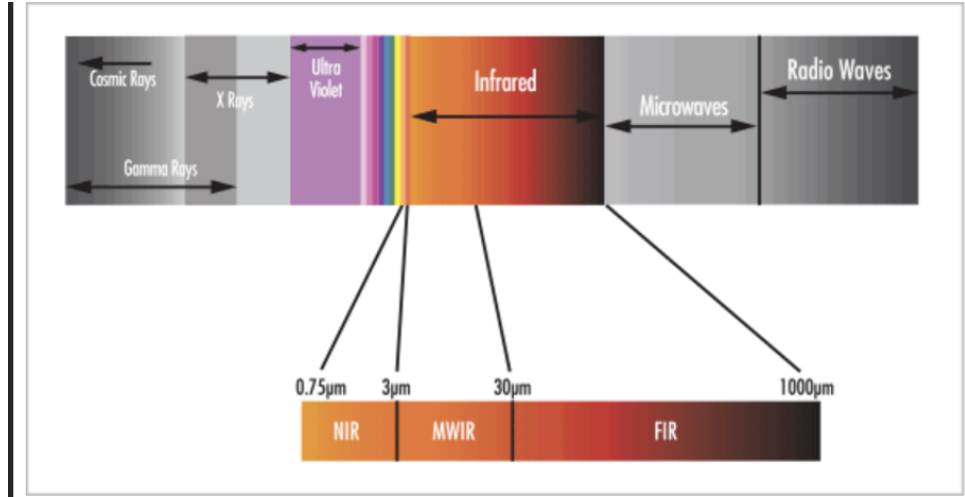


Figure 2: Infrared spectral regions [2]

3 System Expansion

The system requires multiple parallel infrared transmitters. The available digital output resources are:

- 8 pins from PMOD A;
- 8 pins from PMOD B;
- 20 pins from Arduino.

This gives a theoretical total of

$$8 + 8 + 20 = 36$$

available transmitter control pins.

The current system configuration uses 24 transmitters, consisting of all 20 Arduino outputs together with four PMOD A outputs.

Because many transmitters may be located in the same physical region, increasing transmitter range alone is not sufficient. The radiation pattern of each transmitter is also important.

Directional emitters can reduce the amount of optical energy reaching adjacent receiver regions, thereby improving spatial separation between independent channels.

4 Single-Transmitter, 144-Receiver Array Configuration

The experimental receiver system consists of 144 individual infrared receiver units.

Each receiver occupies an approximate footprint of

$$2.5 \text{ cm} \times 2.5 \text{ cm}.$$

The receivers are arranged as a contiguous

$$12 \times 12$$

array.

The dimensions of the complete array are therefore approximately

$$12(2.5 \text{ cm}) = 30 \text{ cm}$$

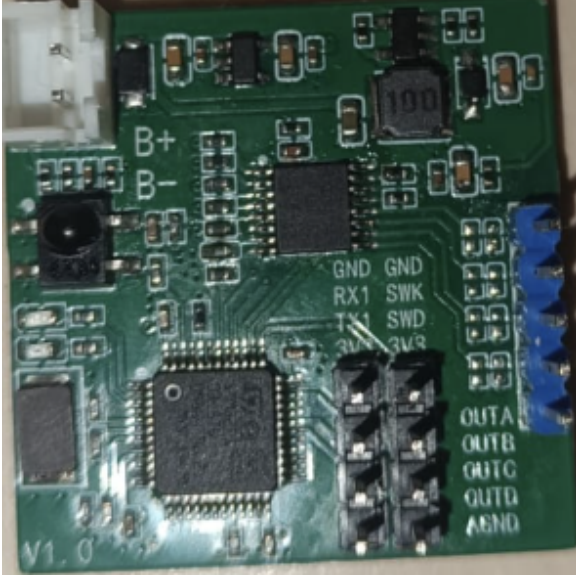
in both directions.

Hence, the receiving surface is approximately

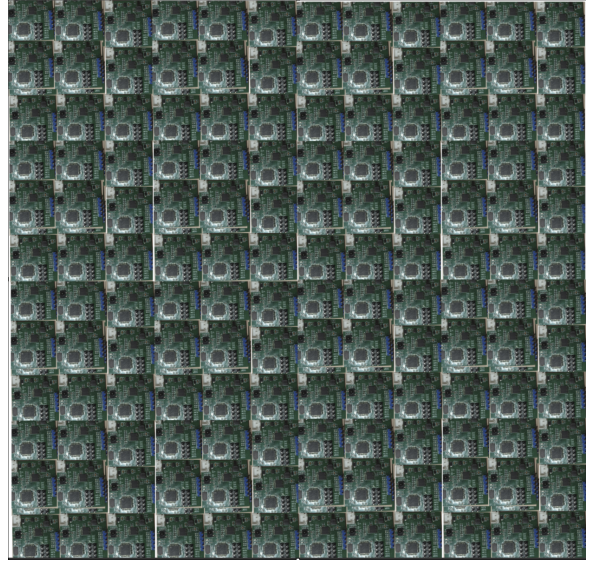
$$30 \text{ cm} \times 30 \text{ cm}$$

with a total area of

$$900 \text{ cm}^2.$$



(a) Single IR receiver unit ($2.5 \times 2.5 \text{ cm}$)



(b) Complete 12×12 receiver grid

Figure 3: Infrared receiver configuration

4.1 Limitations of the KY-005 Transmitter

The initial experimental transmitter was the KY-005 infrared transmitter module [3, 4].

The module contains a conventional 5 mm infrared LED and is intended primarily for short-range microcontroller applications.

Its approximate characteristics are summarized in Table 1.

Table 1: KY-005 Module Specifications

Parameter	Value
Operating voltage	5 V
Forward current	approximately 30–60 mA
Power consumption	approximately 90 mW
Operating temperature	−25°C to 80°C
Dimensions	18.5 × 15 mm

Experimental testing showed that the KY-005 could not provide reliable communication to all locations within the 12 × 12 receiver array at the intended operating distance.

The replacement transmitter must therefore provide greater useful radiant intensity while maintaining an appropriate radiation pattern.

5 Optical Geometry for 1–2 m Transmission

The required beam angle can be estimated directly from the receiver geometry.

The distance between the centre of the 30 × 30 cm array and one of its corners is

$$r = \sqrt{(0.15)^2 + (0.15)^2}$$

which gives

$$r \approx 0.212 \text{ m.}$$

For a transmitter placed a perpendicular distance d from the receiver plane, the angle between the optical axis and one corner is

$$\alpha = \tan^{-1} \left(\frac{0.212}{d} \right).$$

Therefore, the required full angular coverage is approximately

$$\theta_{\text{required}} = 2\alpha = 2 \tan^{-1} \left(\frac{0.212}{d} \right).$$

The calculated values are shown in Table 2.

Table 2: Required Beam Angle for the Receiver Array

Distance	Required Full Angle
1.0 m	24.0°
1.2 m	20.1°
1.5 m	16.1°
2.0 m	12.1°

These calculations show that the required beam is relatively narrow. At a distance of 1 m, approximately 24° of full angular coverage is sufficient to geometrically include the entire receiver surface.

At 2 m, the required angle decreases to approximately 12° .

The transmitter should therefore use a directional IR emitter rather than a very wide radiation pattern. This concentrates a greater fraction of the emitted infrared power toward the intended receiver array and reduces unnecessary optical spill.

6 Infrared Emitter Selection

Two Vishay 940 nm infrared emitters were considered:

- TSAL6200;
- TSAL6100.

Both devices are designed for infrared free-air transmission and have fast optical switching characteristics suitable for a 38 kHz carrier [5, 6].

Their main characteristics are compared in Table 3.

Table 3: Comparison of Candidate 940 nm Infrared Emitters

Parameter	TSAL6200	TSAL6100
Peak wavelength	940 nm	940 nm
Angle of half intensity	$\pm 17^\circ$	$\pm 10^\circ$
Approx. full half-intensity angle	34°	20°
Typical radiant intensity at 100 mA	72 mW/sr	170 mW/sr
Typical forward voltage at 100 mA	1.35 V	1.35 V
Maximum forward current	100 mA	100 mA
Typical rise time	15 ns	15 ns
Typical fall time	15 ns	15 ns

6.1 TSAL6200 as the Primary Emitter

The TSAL6200 is selected as the primary emitter.

Its nominal angle of half intensity is

$$\phi = \pm 17^\circ.$$

The diameter corresponding to this half-intensity cone at distance d is approximately

$$D = 2d \tan(17^\circ).$$

At 1 m,

$$D_{1\text{ m}} = 2(1) \tan(17^\circ) \approx 0.61 \text{ m}.$$

At 2 m,

$$D_{2\text{ m}} = 2(2) \tan(17^\circ) \approx 1.22 \text{ m}.$$

The receiver array has a width of only 0.30 m. Consequently, all four corners of the receiver array lie inside the nominal half-intensity cone at the minimum expected operating distance of approximately 1 m.

This provides useful alignment margin while remaining considerably more directional than a wide-angle infrared emitter.

6.2 TSAL6100 as a Narrow-Beam Alternative

The TSAL6100 has an angle of half intensity of approximately

$$\pm 10^\circ$$

and a typical radiant intensity of approximately

$$170 \text{ mW/sr}$$

at 100 mA.

Its narrower radiation pattern produces greater axial radiant intensity and may therefore be advantageous when longer transmission range or lower optical spill is required.

At a distance of 1 m, the corners of the receiver array are located at approximately

$$\tan^{-1} \left(\frac{0.212}{1} \right) \approx 12^\circ$$

from the optical axis.

The receiver corners therefore lie slightly outside the TSAL6100 nominal $\pm 10^\circ$ half-intensity cone at 1 m.

At approximately 1.2 m and greater distances, the required corner angle becomes close to or smaller than 10° .

For this reason, the TSAL6100 will be treated as an experimental alternative, particularly for longer-distance and low-interference tests.

7 Multi-LED Transmitter Architecture

The transmitter is designed around series-connected pairs of IR LEDs.

The basic configuration contains:

$$2 \times \text{TSAL6200.}$$

The two LEDs are connected in series and therefore carry the same current.

A second identical branch may be populated if additional optical output is required.

The fully populated configuration therefore consists of

$$4 \times \text{TSAL6200}$$

arranged as two parallel branches, with two LEDs in series in each branch.

Each parallel branch has its own current-limiting resistor.

This is necessary because small differences in LED forward voltage can otherwise produce unequal current distribution between parallel LED paths.

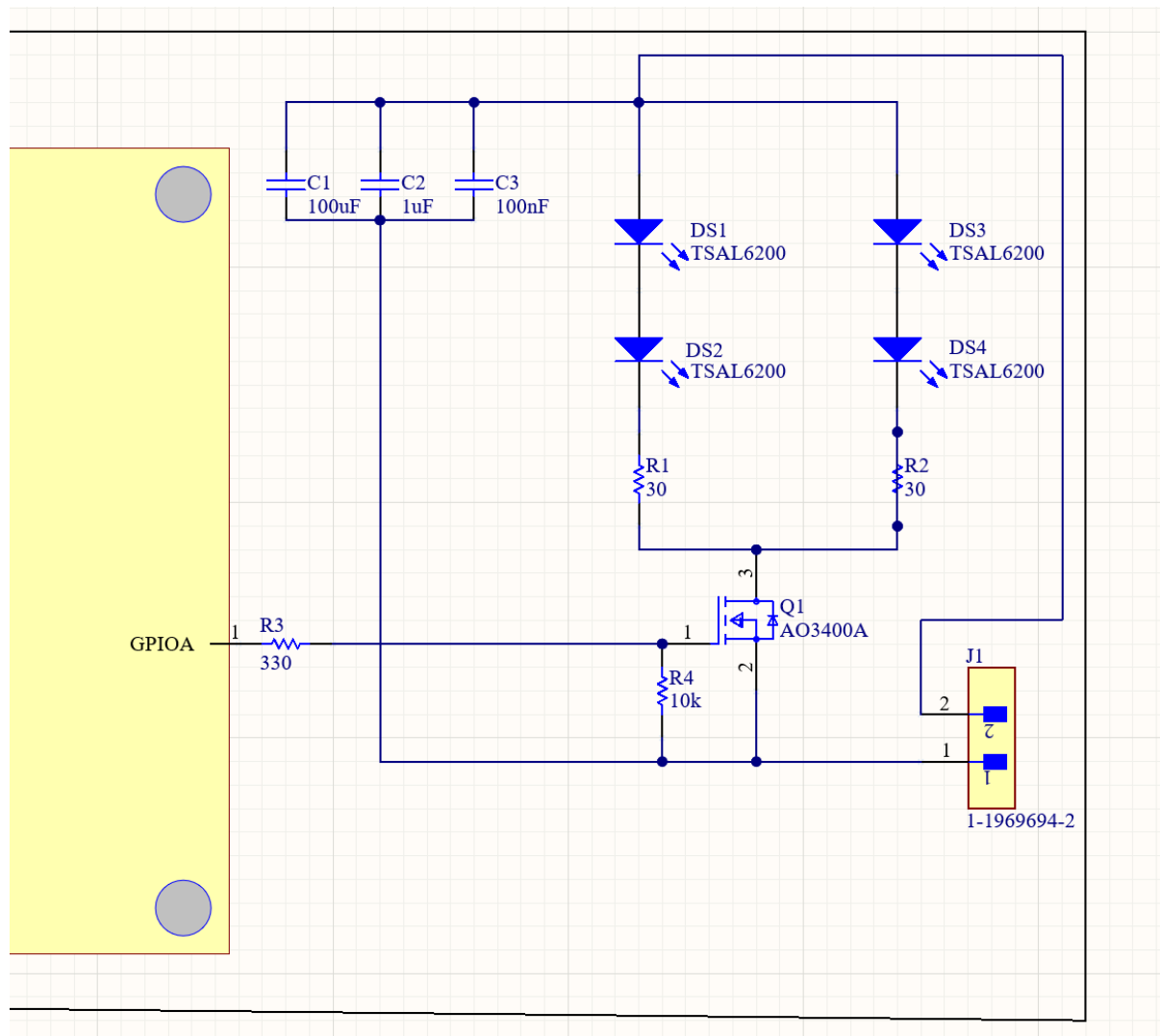
8 MOSFET Driver Circuit

The 3.3 V FPGA output is used only as a control signal. The LED current is supplied from an external regulated 5 V supply.

An AO3400A logic-level N-channel MOSFET is used as the low-side switch [7].

The AO3400A is suitable for direct control from low-voltage digital logic because its drain-source on-resistance is specified at gate voltages as low as 2.5 V.

The proposed circuit is shown in



8.1 Single-Emitter Transmitter Circuit

As an additional experimental configuration, a transmitter channel can use only one TSAL6200 emitter instead of the two-LED series branch. The rest of the driver remains unchanged: the AO3400A is used as the low-side switch, the FPGA drives the gate through a $330\ \Omega$ resistor, and a $10\ \text{k}\Omega$ pull-down resistor keeps the MOSFET OFF during startup.

For a single TSAL6200 on a regulated 5 V supply, the LED current-limiting resistor is increased from $30\,\Omega$ to $47\,\Omega$. Using $V_F \approx 1.35\,\text{V}$,

$$I_F \approx \frac{5 - 1.35}{47} \approx 77.7 \text{ mA}.$$

The corresponding instantaneous resistor dissipation is

$$P_R = I_F^2 R \approx (0.0777)^2 (47) \approx 0.28 \text{ W},$$

therefore a 47Ω , 0.5 W resistor is suitable for the prototype.

This single-emitter version is useful as a minimum-power test configuration. If one LED already provides reliable decoding at the required receiver positions, there is no reason to add unnecessary optical power. A single emitter does not eliminate interference from other independent transmitters, but the lower optical output can reduce unwanted illumination of neighboring receiver regions.

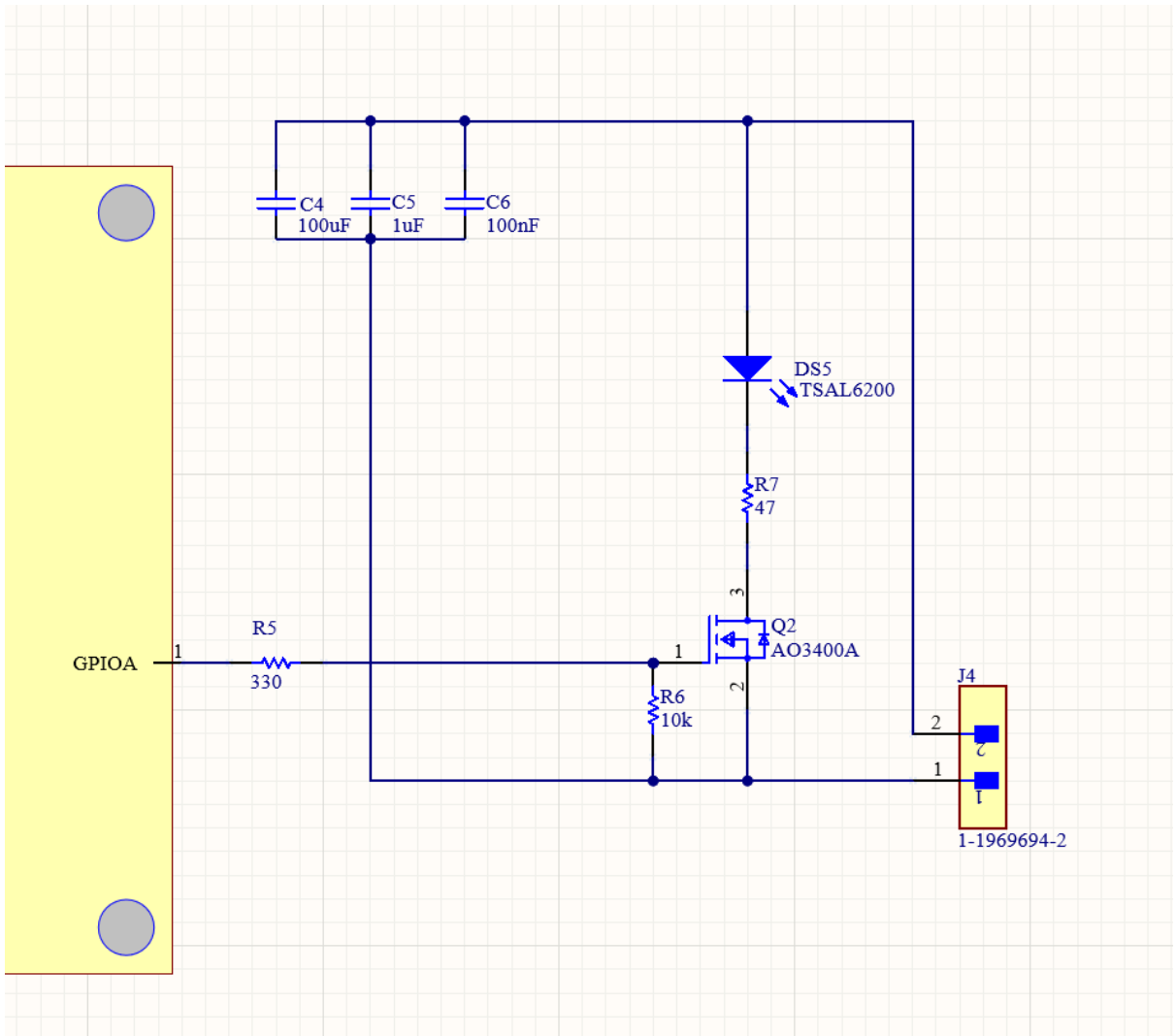


Figure 4: Single-emitter transmitter circuit using one TSAL6200 and a 47Ω current-limiting resistor.

8.2 Circuit Component Specifications

The proposed component values are summarized in Table 4.

Table 4: Transmitter Circuit Components

Component	Value / Part	Purpose
IR emitters	TSAL6200 [5]	940 nm directional optical transmission
Alternative emitter	TSAL6100 [6]	Narrower beam for comparison testing
MOSFET Q_1	AO3400A [7]	Logic-level low-side switch
R_G	330 Ω , 1/4 W [10]	Limits transient FPGA gate charging current
R_{PD}	10 k Ω , 1/4 W [9]	Holds MOSFET OFF during FPGA startup
R_{S1}, R_{S2}	30 Ω , 0.5 W [8]	Sets current independently in each LED branch
Single-emitter test resistor	47 Ω , 0.5 W	Sets approximately 78 mA with one TSAL6200 on 5 V
C_1	100 μ F, 10 V or greater	Local energy storage
C_2	1 μ F ceramic	Local supply decoupling
C_3	100 nF ceramic	High-frequency supply decoupling
Power supply	Regulated 5 V DC	Powers LED branches
Control input	3.3 V FPGA logic	NEC-modulated 38 kHz switching signal

8.3 LED Branch Current Calculation

The typical forward voltage of the TSAL6200 at approximately 100 mA is

$$V_F \approx 1.35 \text{ V}.$$

Because two emitters are connected in series,

$$V_{\text{LED,string}} = 2V_F$$

and therefore

$$V_{\text{LED,string}} = 2(1.35) = 2.70 \text{ V}.$$

The AO3400A voltage drop at the expected current is small compared with the supply voltage, so the nominal branch current can initially be estimated as

$$I_F \approx \frac{V_{\text{supply}} - 2V_F}{R_S}.$$

Using

$$V_{\text{supply}} = 5 \text{ V}$$

and

$$R_S = 30 \Omega,$$

the nominal current becomes

$$I_F \approx \frac{5 - 2(1.35)}{30}$$

which gives

$$I_F \approx 76.7 \text{ mA}.$$

Therefore, the nominal branch current is approximately

$$\boxed{I_F \approx 77 \text{ mA}}.$$

The two LEDs in one series branch both carry this same current.

8.4 Single-Emitter Current Calculation

For the single-emitter configuration shown in Figure 4, only one TSAL6200 is present in the LED current path. To keep approximately the same LED current, the resistor must therefore be changed to 47Ω :

$$I_{F,\text{single}} \approx \frac{5 - 1.35}{47} \approx 77.7 \text{ mA}.$$

Using the original 30Ω resistor with only one LED would instead give approximately

$$\frac{5 - 1.35}{30} \approx 122 \text{ mA},$$

which is above the intended continuous-current operating point used in this design. The single-emitter prototype therefore uses 47Ω , 0.5 W .

8.5 Current with Two LED Branches

When both LED branches are populated, each branch carries approximately

$$77 \text{ mA}.$$

Therefore, when the MOSFET is ON,

$$I_{\text{total}} \approx 2(77 \text{ mA})$$

or

$$\boxed{I_{\text{total}} \approx 154 \text{ mA}}.$$

The actual current should be measured during prototype testing because LED forward voltage varies between devices and with temperature.

8.6 Current-Limiting Resistor Dissipation

The instantaneous power dissipated by each current-limiting resistor is

$$P_R = I^2 R.$$

Using

$$I = 0.077 \text{ A}$$

and

$$R = 30 \Omega,$$

gives

$$P_R = (0.077)^2(30)$$

and therefore

$$P_R \approx 0.178 \text{ W}.$$

A 0.5 W resistor provides sufficient nominal power margin.

Because the LEDs are modulated according to NEC carrier bursts, the long-term average dissipation is lower than the instantaneous value.

8.7 MOSFET Conduction Loss

The AO3400A has a specified maximum

$$R_{\text{DS(on)}} \approx 48 \text{ m}\Omega$$

at

$$V_{\text{GS}} = 2.5 \text{ V}.$$

For the fully populated transmitter,

$$I_D \approx 0.154 \text{ A}.$$

The approximate MOSFET conduction loss is

$$P_Q = I_D^2 R_{\text{DS(on)}}.$$

Therefore,

$$P_Q = (0.154)^2(0.048)$$

which gives approximately

$$P_Q \approx 1.1 \text{ mW}.$$

The MOSFET conduction loss is therefore very small for the expected operating current.

8.8 Switching-Speed Requirement

The NEC carrier period is approximately

$$T_c = 26.3 \mu s.$$

The TSAL6200 has typical optical rise and fall times of approximately

$$t_r \approx 15 \text{ ns}$$

and

$$t_f \approx 15 \text{ ns}.$$

The ratio between the optical rise time and carrier period is

$$\frac{15 \text{ ns}}{26.3 \mu s} \approx 5.7 \times 10^{-4}.$$

The emitter switching time is therefore negligible compared with the 38 kHz carrier period.

For this application, optical coverage and radiant intensity are more significant design parameters than LED switching speed.

8.9 Connection Instructions

The transmitter should be connected as follows:

1. Connect the external regulated +5 V supply to the first IR LED in each populated branch.
2. Connect two TSAL6200 emitters in series in each branch.
3. Install a separate 30 Ω , 0.5 W resistor in each branch.
4. Connect the lower ends of the two branches to the drain of the AO3400A.
5. Connect the AO3400A source directly to ground.
6. Connect the FPGA output to the MOSFET gate through a 330 Ω resistor.
7. Connect a 10 k Ω pull-down resistor between the MOSFET gate and ground.
8. Connect the FPGA ground and the external 5 V power-supply ground together.
9. Place the 100 μ F, 1 μ F, and 100 nF bypass capacitors between +5 V and ground close to the LED branches and MOSFET.

8.10 Operating Principle

The FPGA generates the NEC waveform containing the 38 kHz carrier.

When the FPGA output is HIGH, the AO3400A gate is driven to approximately 3.3 V relative to its grounded source.

The MOSFET turns ON, allowing current to flow through each populated LED branch:

$$+5\text{ V} \rightarrow LED \rightarrow LED \rightarrow R_S \rightarrow Q_1 \rightarrow \text{GND}.$$

The infrared emitters therefore generate the optical carrier.

When the FPGA output becomes LOW, the MOSFET turns OFF and current through the LED branches stops.

The 10 k Ω gate pull-down resistor ensures that the MOSFET remains OFF when the FPGA output is unconfigured, such as during startup or reset.

The gate resistor reduces the short transient current required to charge and discharge the MOSFET gate capacitance.

8.11 Power-Supply Decoupling

Fast switching of the IR LED current produces transient changes in supply current.

Local bypass capacitors are therefore placed near the transmitter circuit.

The proposed capacitor network contains:

$$C_1 = 100\text{ }\mu\text{F},$$

$$C_2 = 1\text{ }\mu\text{F},$$

and

$$C_3 = 100\text{ nF}.$$

The larger capacitor provides local energy storage while the smaller ceramic capacitors suppress higher-frequency supply disturbances.

PCB traces between these capacitors, the LED branches, and the MOSFET should be kept short.

8.12 Power-Supply Requirement

A single fully populated transmitter is expected to draw approximately

$$154\text{ mA}$$

during the ON portion of the carrier waveform.

A regulated 5 V, 1 A supply therefore provides substantial current margin for a single experimental transmitter.

When multiple transmitters share a common supply, the required supply current depends on the maximum number of simultaneously active transmitters.

For example, if all 24 transmitters were fully populated and switched ON simultaneously, the instantaneous LED current would be approximately

$$I_{\text{system}} = 24(0.154)$$

which gives

$$I_{\text{system}} \approx 3.7 \text{ A.}$$

This value represents an instantaneous upper-bound estimate for the LED transmitter circuits only.

A shared supply should therefore include additional margin for the control electronics, wiring losses, transient current, and other system loads.

8.13 Thermal Considerations

The emitter branches operate below the 100 mA forward-current rating of the selected IR LEDs.

The current-limiting resistors dissipate approximately 0.18 W during the ON state, while the calculated AO3400A conduction loss is only on the order of 1 mW.

The NEC waveform additionally reduces average dissipation because the LEDs are not continuously energized.

Normal PCB spacing and ventilation are therefore expected to be adequate for the prototype.

Component temperature should nevertheless be checked during initial continuous testing.

9 Multi-Transmitter Interference

Multiple LEDs connected to the same switching circuit and carrying the same NEC signal behave as one optical transmitter. Adding LEDs in this case increases the available optical output without introducing a new data signal.

A different situation occurs when independent transmitters send different NEC messages simultaneously.

Because the transmitters use the same 940 nm optical region and the same 38 kHz carrier, radiation from more than one transmitter may reach the same receiver.

The receiver cannot identify the physical origin of overlapping 38 kHz optical signals solely from their carrier frequency. Simultaneous signals may therefore corrupt the received NEC waveform.

For this reason, transmitter directivity is an important part of the system design.

The proposed directional emitters reduce interference by concentrating the radiation toward the intended receiver area.

9.1 Spatial Isolation

Spatial isolation can be improved through several methods:

- selecting an appropriate emitter beam angle;
- physically separating adjacent transmitters;
- carefully orienting each transmitter toward its intended receiver region;

- using opaque tubes or barriers around the emitters;
- reducing optical output to the minimum level required for reliable communication.

Using the minimum necessary optical output is particularly important. Once reliable decoding is achieved at the worst-position receiver, additional optical power may increase interference without improving the useful communication link.

10 Prototype and Experimental Procedure

The transmitter PCB should support four IR LED positions arranged as two independently current-limited series branches.

The initial prototype should populate only one branch containing two TSAL6200 emitters.

The following measurements should then be performed:

1. First test the single-emitter configuration using one TSAL6200, a $47\ \Omega$, 0.5 W resistor, and approximately 78 mA LED current. If this configuration already gives reliable reception at the required receiver positions, record it as the minimum-power operating point.
2. Operate two TSAL6200 emitters at approximately 77 mA.
3. Position the transmitter approximately 1.0 m from the receiver array.
4. Verify NEC reception at the centre of the array.
5. Verify reception at the four edge-centre positions.
6. Verify reception at the four corners of the array.
7. Repeat the measurements at approximately 1.5 m.
8. Repeat the measurements at approximately 2.0 m.
9. If reception margin is insufficient, populate the second two-LED branch.
10. Repeat the complete distance and position measurements using four emitters.
11. Compare the TSAL6200 configuration with the narrower-beam TSAL6100 configuration.

10.1 Receiver Coverage Criterion

The transmitter should not be evaluated only at the centre of the receiver grid.

Reliable operation should be verified at the positions where optical intensity is expected to be lowest, particularly the four corners.

The transmitter will be considered to provide sufficient optical coverage if all tested receiver positions can decode the NEC signal reliably at the required operating distance.

The desired operating point is therefore the lowest optical output that provides adequate decoding margin at the worst-case receiver location.

10.2 Interference Test

After single-transmitter operation has been validated, two adjacent transmitters should be tested together.

Testing should include:

1. only transmitter A active;
2. only transmitter B active;
3. A and B transmitting at different times;
4. A and B transmitting simultaneously;
5. different physical separation between the transmitters;
6. different transmitter orientations;
7. optional optical barriers or guiding tubes.

The experiment should determine whether narrower radiation patterns or physical isolation significantly reduce cross-channel interference.

10.3 Beam-Angle Comparison Experiment

The TSAL6200 and TSAL6100 provide an opportunity to experimentally study the trade-off between beam width and radiant intensity.

The TSAL6200 provides a wider

$$\pm 17^\circ$$

half-intensity angle, whereas the TSAL6100 provides a narrower

$$\pm 10^\circ$$

half-intensity angle.

The narrower emitter produces greater axial radiant intensity but is more sensitive to alignment.

For each emitter type, the following should be measured:

- maximum reliable communication distance;
- reliability at receiver-array corners;
- tolerance to lateral transmitter misalignment;
- interference observed at neighboring receivers.

These measurements will determine which radiation pattern provides the best compromise between receiver coverage and transmitter isolation.

11 PCB Design Considerations

The transmitter PCB should provide four emitter footprints.

The emitters should be positioned symmetrically around the optical centre of the transmitter so that multiple LEDs illuminate approximately the same receiver region.

The two LED branches should have identical trace geometry where practical.

The AO3400A and bypass capacitors should be positioned close to the LED branches to minimize switching-loop area.

The ground path between the MOSFET and power supply should have low impedance.

The FPGA control trace does not carry the LED current and may therefore be routed separately from the power path.

Where transmitter interference is significant, mechanical mounting should permit adjustment of the emitter orientation.

12 Proposed Transmitter Configuration

The baseline transmitter configuration is summarized in Table 5.

Table 5: Proposed IR Transmitter Configuration

Parameter	Design Value
IR wavelength	940 nm
Protocol	NEC
Carrier frequency	38 kHz
Operating distance	approximately 1–2 m
Receiver area	30×30 cm
Primary emitter	Vishay TSAL6200
Emitter angle of half intensity	$\pm 17^\circ$
Initial emitter count	2
Maximum prototype emitter count	4
LED arrangement	2 LEDs per series branch
Number of branches	1 initially, 2 optional
Nominal branch current	approximately 77 mA
MOSFET	AO3400A
Supply voltage	5 V
FPGA control voltage	3.3 V
Branch resistor	30Ω , 0.5 W
Single-emitter test configuration	1 TSAL6200 with 47Ω , 0.5 W
Gate resistor	330Ω
Gate pull-down	10 k Ω

13 Conclusion

A directional 940 nm infrared transmitter has been proposed for communication with a 12×12 receiver array located approximately 1–2 m from the transmitter.

Geometrical analysis shows that the complete 30×30 cm receiver array requires approximately 24° of full angular coverage at 1 m and only approximately 12° at 2 m.

The Vishay TSAL6200, with an angle of half intensity of approximately $\pm 17^\circ$, provides suitable coverage margin for this distance range while maintaining substantially greater directionality than a very wide-beam emitter.

Two TSAL6200 emitters connected in series form the baseline configuration. A second identical branch allows the transmitter to be expanded to four emitters if experimental measurements show that additional optical output is required.

The emitters are controlled using an AO3400A N-channel MOSFET driven by the FPGA's 3.3 V NEC signal. Each LED branch includes its own current-limiting resistor and operates at a nominal current of approximately 77 mA.

The final number of emitters should be determined experimentally by testing the centre, edges, and corners of the receiver array at 1.0 m, 1.5 m, and 2.0 m.

A single-emitter TSAL6200 configuration with a $47\ \Omega$, 0.5 W resistor is also included as a minimum-power test case. If it provides reliable decoding at the required distance, it can reduce unnecessary optical spill compared with the higher-output configurations.

The same test platform can also compare the wider TSAL6200 with the narrower TSAL6100 and quantify the trade-off between receiver coverage, radiant intensity, alignment tolerance, and interference between neighboring transmitters.

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